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SYSTEMS OF LINEAR ALGEBRAIC EQUATIONS

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CONTENT

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Genki

1. Introduction

Linear Algebraic Equations

Focus on solving systems of linear algebraic equations of the form

$$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1$$

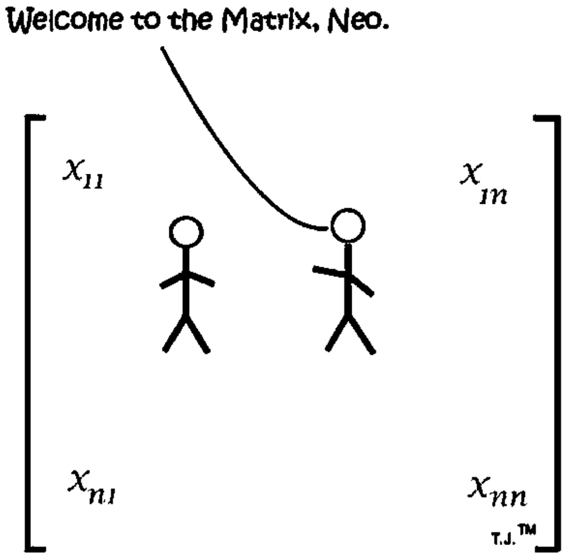
$$a_{n1}x_1 + a_{n2}x_2 + \ldots + a_{nn}x_n = b_n$$

- **Motivation**

1. Transition from solving a single equation to systems of n equations in n unknowns.
2. Distinction between linear and nonlinear systems; focus here on linear.
3. Historical difficulty of hand-solving large systems ($n \geq 4$) and the enabling role of computers.
4. Emphasis on using computational power to enhance creativity in problem formulation and interpretation.

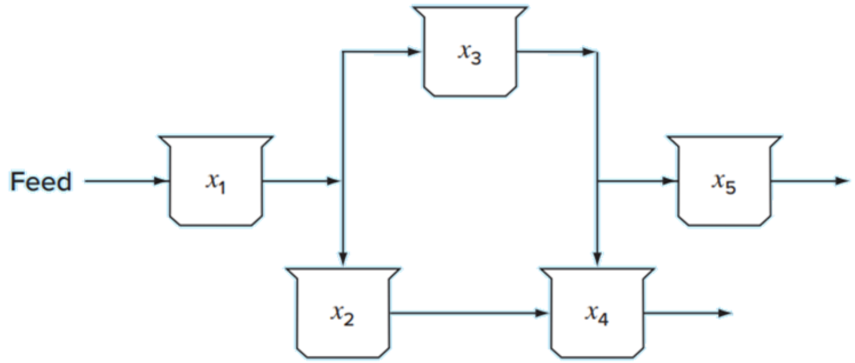
Noncomputer Methods for Solving Systems of Equations

- Simple techniques (elimination, substitution) work for small systems ($n \leq 3$).
- For larger systems, hand methods become impractical; computers are required.

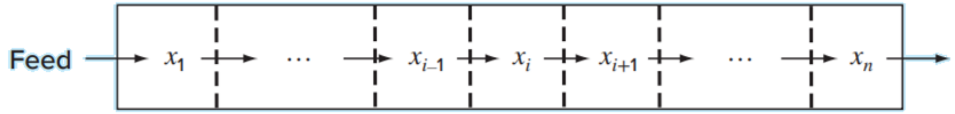


Linear Algebraic Equations and Engineering Practice

- Many engineering models arise from conservation laws (mass, energy, momentum).
- Example: mass balance in a series of chemical reactors (lumped-parameter system).
- Distinction between lumped (finite components) and distributed (continuum) models.
- Coupling of variables: each equation's unknown can depend on multiple components.
- Other applications include regression analysis and spline interpolation.



Lumped variable system that involves coupled finite components



Distributed variable system that involves a continuum

Mathematical Background

Matrix Notation

- Definition of an n×m matrix [A] with elements a_{ij}; rows (i) and columns (j).
- Row vectors (1×m) and column vectors (n×1); square matrices (n=m).

$$[A] = \begin{array}{cc} \begin{matrix} \phantom{a_{11}} & \phantom{a_{12}} & \text{Column 3} & & \phantom{a_{1m}} \end{matrix} & \\ \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1m} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2m} \\ \cdot & \cdot & & & \cdot \\ \cdot & \cdot & & & \cdot \\ \cdot & \cdot & & & \cdot \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nm} \end{bmatrix} & \begin{matrix} \\ \text{Row 2} \\ \\ \\ \\ \\ \end{matrix} \end{array}$$

Mathematical Background

Special Types of Square Matrices

- Symmetric ($a_{ij} = a_{ji}$)
- Diagonal (off-diagonal entries zero)
- Identity (diagonal entries = 1)
- Upper and lower triangular

Mathematical Background

Matrix Operating Rules

- Equality, addition, subtraction (commutative, associative).
- Scalar multiplication.
- Matrix multiplication ($[A]_{n \times m} \cdot [B]_{m \times l} = [C]_{n \times l}$), non-commutative.
- Transpose ($[A]^T$), trace (sum of diagonal), inverse ($[A]^{-1}$ when $[A]$ is square & nonsingular), augmentation (adding columns).

Organization

- Small-system methods, naïve Gauss elimination, pivoting, Gauss-Jordan.
- LU decomposition, matrix inversion, condition number.
- Engineering case studies (chemical, civil, electrical, mechanical).

2. Gauss Elimination

Overview

- Deals with solving simultaneous linear algebraic equations of the form

$$a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n = b_1$$
$$a_{n1}x_1 + \ldots + a_{nn}x_n = b_n$$

- Called "Gauss elimination" because it combines equations to eliminate unknowns.

- Fundamental algorithm in modern software for linear equation solving.

$$\begin{matrix} 1 & 0 & 4 & 2 \\ 1 & 2 & 6 & 2 \\ 2 & 0 & 8 & 8 \\ 2 & 1 & 9 & 4 \end{matrix}$$

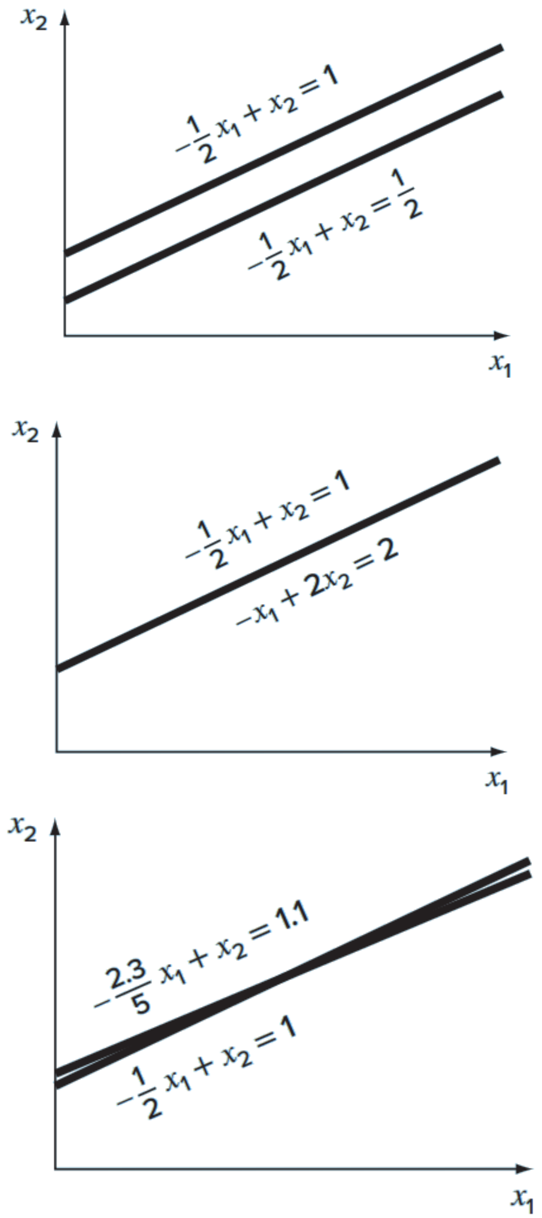
Solving Small Numbers of Equations ($n \leq 3$)

Three classical methods that do not require a computer:

1. **Graphical Method**
2. **Cramer's Rule**
3. **Elimination of Unknowns**

Graphical Method

- For two equations: plot each as a straight line on Cartesian axes ($x_1\ vs.\ x_2$).
- Solve for x_2 in each equation to get slope–intercept form ($x_2 \ = \ m\, x_1 \ + \ c$).
- Intersection of the two lines gives the solution (x_1, x_2).
- For three equations: represent each as a plane in 3D; solution is the common intersection point.
- Limitations:
  - Parallel lines/planes → no solution.
  - Coincident lines/planes → infinitely many solutions (singular systems).
  - Nearly coincident → ill-conditioned, highly sensitive to round-off error.



Determinants and Cramer's Rule

Determinant (D): a scalar value computed from the coefficients matrix $[A]$

- 2×2: D = $a_{11}a_{22} - a_{12}a_{21}$
- 3×3: expansion by minors, e.g.

$$D = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$[A] = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

- Cramer's Rule: each unknown x_i is

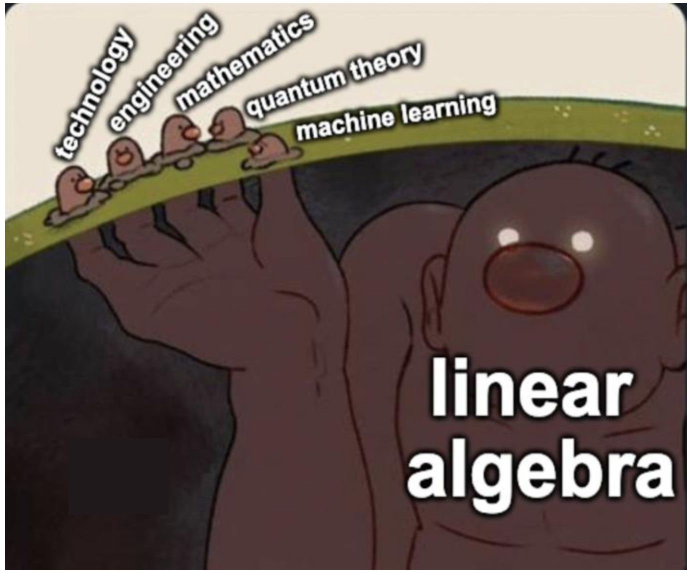
$$x_i = \frac{\det([A] with\ column\ i\ replaced\ by\ \{b\})}{D}$$

- Practical for small systems but impractical for more than three equations due to computational cost.

Elimination of Unknowns

- Algebraic method: multiply equations by appropriate constants and subtract to eliminate one unknown.

- For larger systems, the same principle applies but becomes tedious by hand; forms the basis for the Gauss elimination algorithm in computers.



Example 1

- Use Cramer's rule to solve

$$
\begin{aligned}
0.3x_1 + 0.52x_2 + x_3 &= -0.01 \\
0.5x_1 + x_2 + 1.9x_3 &= 0.67 \\
0.1x_1 + 0.3x_2 + 0.5x_3 &= -0.44
\end{aligned}
$$

Overview of Naïve Gauss Elimination

- A direct method to solve a general system of n linear equations in n unknowns.
- Consists of two main phases: forward elimination (to produce an upper-triangular system) and back substitution (to compute the unknowns).

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right]$$

$$\Downarrow$$

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ & a'_{22} & a'_{23} & b'_2 \\ & & a''_{33} & b''_3 \end{array}\right]$$

Forward elimination

$$\Downarrow$$

$$\begin{aligned} x_3 &= b''_3/a''_{33} \\ x_2 &= (b'_2 - a'_{23}x_3)/a'_{22} \\ x_1 &= (b_1 - a_{12}x_2 - a_{13}x_3)/a_{11} \end{aligned}$$

Back substitution

Overview of Naïve Gauss Elimination

- **Forward Elimination**

1. **Pivot selection**: for each row i=1,2,…,n−1, use equation i as the pivot equation, with pivot element a_{ii}.
2. **Elimination**: for each lower row k=i+1,…,n:
   - Compute the multiplier

$$m_{k,i} = \frac{a_{k,i}}{a_{i,i}}$$

   - Update coefficients for j=i,…,n:

$$a_{k,j} \leftarrow a_{k,j} - m_{k,i}a_{i,j} \qquad\qquad b_k \leftarrow b_k - m_{k,i}b_i$$

3.Result: an upper-triangular system

$$a_{11}x_1 + \cdots + a_{1n}x_n = b_1$$
$$a'_{22}x_2 + \cdots + a'_{2n}x_n = b'_2$$

Overview of Naïve Gauss Elimination

Back Substitution

1.Solve for the last unknown:

$$x_n = \frac{b_n^{n-1}}{a_{nn}^{n-1}}$$

2. For i = n−1, n−2, …,1:

$$x_i = \frac{b_i^{i-1} - \sum_{j=i+1}^{n} a_{ij}^{i-1} x_j}{a_{ii}^{i-1}}$$

Example 2

- Use Gauss elimination to solve

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85$$
$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3$$
$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4$$

Carry six significant figures during the computation.

Gauss-Jordan Method

Variant of Gaussian Elimination

- Gauss–Jordan is a **variation** of Gaussian elimination.
- Unlike standard Gaussian elimination, when an unknown is eliminated it is removed from *all* equations, not just the subsequent ones.

$$Gauss - Jordan$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & x_1 \\ 0 & 1 & 0 & x_2 \\ 0 & 0 & 1 & x_3 \end{array}\right]$$

Row Normalization

- Every row **is normalized** by dividing by its pivot element.
- This yields an **identity matrix** directly.

Gauss-Jordan Method

No Back Substitution Needed

- Since the elimination produces the **identity matrix**, **back substitution is unnecessary** to obtain the solution.

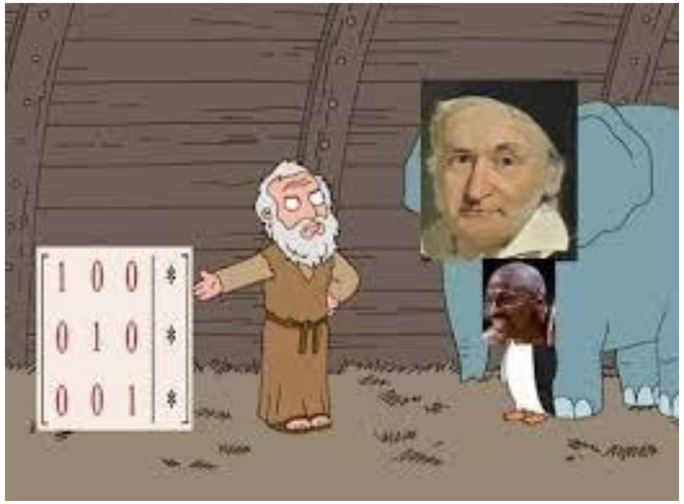
Pivoting and Stability

- All the pitfalls and improvements for Gaussian elimination (e.g., avoiding division by zero, reducing round-off error) apply equally to Gauss–Jordan.
- A similar pivoting strategy can be employed to improve numerical stability.

Gauss-Jordan Method

Practical Preference

- Due to its higher operation count, **Gaussian elimination is generally preferred for solving linear systems.**
- However, Gauss–Jordan remains in use within engineering applications and certain numerical algorithms.



Example 3

Use the Gauss-Jordan technique to solve the same system

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85$$
$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3$$
$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4$$

3. Gauss-Seidel Method

Gauss–Seidel Definition

Provide an alternative to direct elimination by guessing a solution and refining it systematically.

- The most commonly used iterative method for solving

$$[A]\{X\}=\{B\}$$

where [A] is an n×n matrix and {X}, {B} are vectors.

3×3 System Formulation

Assuming all diagonal entries $a11, a22, a33$ are nonzero, solve each equation for one variable:

$$x_1 = \frac{b_1 - a_{12}x_2 - a_{13}x_3}{a_{11}}$$

$$x_2 = \frac{b_2 - a_{21}x_1 - a_{23}x_3}{a_{22}}$$

$$x_3 = \frac{b_3 - a_{31}x_1 - a_{32}x_2}{a_{33}}$$

$$\left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array}\right]$$

Iteration Procedure

- **Step 1:** Use $\{X\}^{j-1}$ (previous iterate) to compute x_1^j.
- **Step 2:** Substitute x_1^j and previous guesses to get x_2^j.
- **Step 3:** Use updated x_1^j, x_2^j to compute x_3^j.
- **Repeat** these three updates in sequence until convergence.

Example 4

Use the Gauss-Seidel method to obtain the solution of the system

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85$$
$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3$$
$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4$$

4. Linear Algebraic Equations with Software Packages

Excel

Two solution methods

- Use the Solver tool.
- Use matrix inversion and multiplication functions.

Analytical formula

$$\{X\} = [A]^{-1}\{B\}$$

Excel implementation

- Built-in function **MINVERSE** to compute $[A]^{-1}$.
- Built-in function **MMULT** to multiply $[A]^{-1}$ by {B}.

Example 5

Use Excel to obtain this solution

$$\begin{bmatrix} 1 & 1/2 & 1/3 \\ 1 & 2/3 & 1/2 \\ 1 & 3/4 & 3/5 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 1.833333 \\ 2.166667 \\ 2.35 \end{Bmatrix}$$

MATLAB

- **Name and Acronym**: MATLAB stands for “MAtrix LABoratory.”
- **Primary Purpose**: Designed specifically to facilitate matrix manipulations.
- **Core Strength**: Offers excellent capabilities for matrix operations.

MATLAB

Matrix Analysis		Linear Equations	
Function	**Description**	**Function**	**Description**
cond	Matrix condition number	\ and /	Linear equation solution; use "help slash"
norm	Matrix or vector norm	chol	Cholesky factorization
rcond	LINPACK reciprocal condition estimator	lu	Factors from Gauss elimination
rank	Number of linearly independent rows or columns	inv	Matrix inverse
det	Determinant	qr	Orthogonal-triangular decomposition
trace	Sum of diagonal elements	qrdelete	Delete a column from the QR factorization
null	Null space	qrinsert	Insert a column in the QR factorization
orth	Orthogonalization	nnls	Nonnegative least squares
rref	Reduced row echelon form	pinv	Pseudoinverse
		lscov	Least squares in the presence of known covariance

Example 6

Use MATLAB to obtain this solution

$$\begin{bmatrix} 1 & 1/2 & 1/3 \\ 1 & 2/3 & 1/2 \\ 1 & 3/4 & 3/5 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 1.833333 \\ 2.166667 \\ 2.35 \end{Bmatrix}$$

5. Case Studies: Application of Engineering

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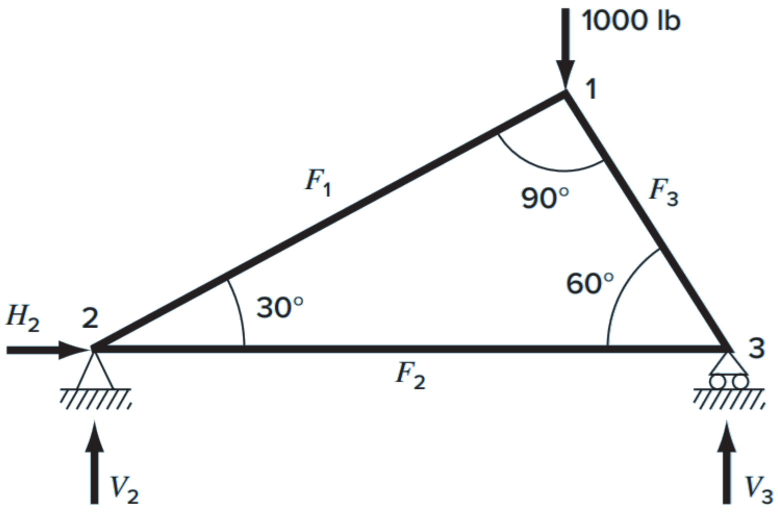
Example 7

An important problem in structural engineering is determining the internal forces and support reactions in a statically determinate truss. Consider the truss shown, composed of three members connected at nodes 1, 2 and 3.

The member forces F_1, F_2 and F_3 may act in either tension or compression. External reactions H_2, V_2 at the pinned support (node 2) and V_3 at the roller support (node 3) characterize how the truss interacts with its supports. A downward load of 1000 lb is applied at node 1.

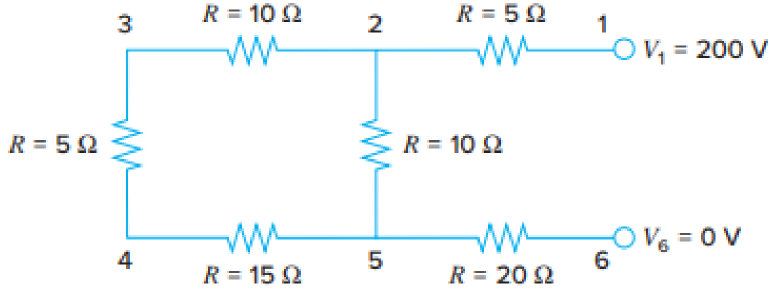
Determine

- The internal axial forces F_1, F_2 and F_3 in each member, and
- The support reactions H_2, V_2 and V_3



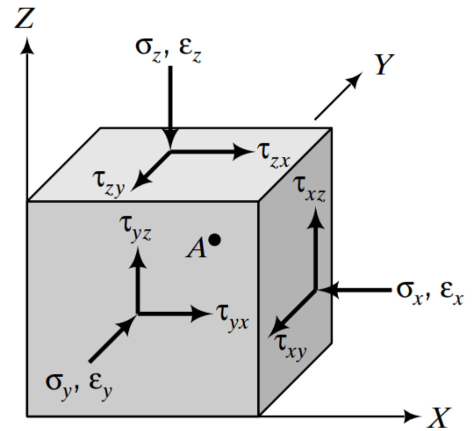
Example 8

Consider the resistor circuit shown in the figure, which contains six branch currents $i_{12}, i_{52}, i_{32}, i_{65}, i_{54} and\ i_{43}$ whose magnitudes and directions are unknown. Using Kirchhoff's current law at each node and Kirchhoff's voltage law around each of the two independent loops, formulate the system of linear equations that must be solved to find all six branch currents. All resistances and the single 200 V source in the circuit are as indicated in the figure.



Example 9

A geotechnical engineer must evaluate the elastic behavior of a saturated clay layer (E = 25 MPa, ν = 0.30) beneath the foundation of a square plate that transmits deformations measured in the field: ε_x = 0.15%, ε_y = 0.10%, ε_z = –0.05%, γ_{xy} = 0.12%, γ_{yz} = 0.08 %, and γ_{zx} = 0.10 %. Determine the normal stresses σ_x, σ_y, σ_z and the shear stresses $\tau_{xy}, \tau_{yz}, \tau_{zx}$ at the base of the foundation, accounting for the theory of isotropic elasticity.



$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_z \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{Bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu & 0 & 0 & 0 \\ -\nu & 1 & -\nu & 0 & 0 & 0 \\ -\nu & -\nu & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2(1+\nu) & 0 & 0 \\ 0 & 0 & 0 & 0 & 2(1+\nu) & 0 \\ 0 & 0 & 0 & 0 & 0 & 2(1+\nu) \end{bmatrix} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{zx} \end{Bmatrix}$$

6. Summary

Summary

Covers fundamental techniques for solving systems of linear algebraic equations, including theory, algorithms, software tools, and engineering applications .

Introduction & Motivation

- Definition of a linear system (n equations in n unknowns).
- Importance in engineering (mass, energy, momentum balances; regression; interpolation).
- Historical challenges of hand-solving large systems and the role of computers .

Non-Computer Methods ($n \leq 3$)

- Elimination and substitution by hand.
- Graphical method, Cramer’s Rule, and manual elimination for very small systems .

Summary

Gauss Elimination

- Forward elimination to upper-triangular form plus back-substitution.
- Pivoting strategies for numerical stability.
- Worked examples demonstrating six-figure accuracy .

Gauss–Jordan Method

- Extension of Gaussian elimination that produces the identity matrix directly.
- Normalizes each row by its pivot; eliminates need for back-substitution.
- Comparison of computational cost versus direct Gaussian elimination.

Gauss–Seidel Iterative Method

- Iterative refinement starting from an initial guess.
- Particularly useful for large sparse systems where direct methods are costly.
- Step-by-step iteration procedure and convergence criteria

Summary

Method	Stability	Precision	Breadth of Application	Programming Effort	Comments
Graphical	—	Poor	Limited	—	May take more time than the numerical method but can be useful for visualization
Cramer's rule	—	Affected by round-off error	Limited	—	Excessive computational effort required for more than three equations
Gauss elimination (with partial pivoting)	—	Affected by round-off error	General	Moderate	
Gauss-Seidel	May not converge if system is not diagonally dominant	Excellent	Appropriate only for diagonally dominant systems	Easy	